

Per-Fire Threshold Calibration Limits Burn Severity Mapping on Unseen Wildfires

Jessie Dong
Stanford University
jessiedong@stanford.edu

Abstract

Burn severity maps determine where post-fire erosion control, debris-flow warnings, and reforestation are deployed. In the United States, the Monitoring Trends in Burn Severity (MTBS) program produces these maps by thresholding the differenced Normalized Burn Ratio (dNBR), with thresholds chosen by an analyst for each fire. Automatic methods are evaluated here on how closely they reproduce these maps on fires excluded from training. The dataset contains 31 California and Oregon wildfires from 2018 to 2021 (7.1 million acres, 31.4 million labeled pixels) on the exact Landsat scenes MTBS used. Five-fold cross-validation holds out whole fires together with the fires that overlap them. The methods are dNBR with generic thresholds, dNBR, RdNBR, and RBR with thresholds learned on the training fires, a per-pixel gradient-boosting classifier, and two U-Net variants. Learned thresholds raise mean mIoU from 0.564 to 0.684 and improve 29 of 31 fires. Neither U-Net improves on calibrated dNBR. The better variant reaches 0.686, a paired difference of +0.002 (95% CI -0.009 to $+0.014$). Each fire’s own analyst thresholds reach 0.735, and their advantage on a fire grows with the distance between its analyst thresholds and the learned ones (Spearman $\rho = 0.68$). Of the remaining errors, 99.8% confuse adjacent classes and 93.2% lie within 30 m of a class transition in the MTBS map. The remaining error is therefore fire-specific calibration rather than missing spectral or spatial information.

1 Introduction

After a large wildfire, land managers need a map of how severely the fire altered vegetation and soil. The map guides where crews place erosion control, which watersheds face elevated debris-flow risk in the first rains, and which stands require replanting rather than natural regeneration. In the United States, the Monitoring Trends in Burn Severity (MTBS) program produces these maps for every fire larger than 1,000 acres in the West [2]. An MTBS analyst selects a pre-fire and a post-fire Landsat scene, computes the differenced Normalized Burn Ratio (dNBR) [5], and chooses dNBR thresholds that separate unburned, low, moderate, and high severity for that fire. For many forested fires, MTBS uses a post-fire scene from the next growing season, which it calls an extended assessment. The resulting maps are the reference product for burn severity in the United States, but each requires an analyst’s judgment.

The analyst’s thresholds differ from fire to fire. Across the 31 fires in the dataset, the threshold between moderate and high severity ranges from 480 to 685 dNBR units, and every value exceeds the generic threshold of 440 that is often used in its place (Figure 2). A method that reproduces MTBS maps automatically must therefore either estimate a fire’s thresholds from its imagery or learn a mapping from imagery to severity that does not depend on them. Deep segmentation models take the second approach. They use all six reflective bands rather than one index, and they use spatial context.

Three choices make the comparison between these approaches informative. First, every method is scored against MTBS analyst maps rather than against labels derived from the same imagery. Labels computed from dNBR with fixed thresholds are a closed-form function of the model’s inputs, and a model trained on them can at best recover that function. Second, the inputs are the exact Landsat scenes named in each fire’s MTBS record, warped onto the 30 m grid of the MTBS mosaic, so labels are never resampled and inputs match what the analyst saw. Third, whole fires are held out. Fires that overlap spatially share a group, and

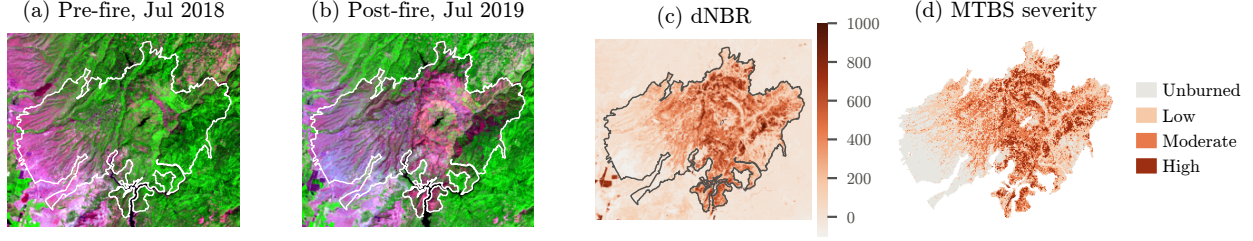


Figure 1: The 2018 Camp Fire. (a, b) Landsat surface reflectance before and after the fire (SWIR2, NIR, and red composite), from the scenes named in the MTBS record. (c) dNBR computed from the two scenes. (d) The MTBS severity map, produced by an analyst who chose dNBR thresholds for this fire. The white or gray line is the MTBS fire perimeter, and only pixels inside it are labeled. A method receives (a) and (b) and predicts (d).

groups are assigned to five cross-validation folds, so every fire is predicted by a model that has seen neither that fire nor any fire that overlaps it.

Under these conditions, threshold calibration dominates. Replacing the generic dNBR thresholds with thresholds learned on the training fires raises mean mIoU by +0.121 (95% CI +0.094 to +0.146). Two U-Net variants, one on the twelve reflectance bands and one with five additional index channels, match calibrated dNBR but do not exceed it, with paired differences of -0.001 and $+0.002$ and intervals that include zero. An oracle that applies each test fire’s own analyst thresholds reaches 0.735, and its advantage on a fire increases with the distance between that fire’s analyst thresholds and the learned ones. Nearly all remaining errors confuse adjacent severity classes, and they concentrate at transitions between classes in the MTBS map. Labels derived by applying generic thresholds to dNBR also depend strongly on preprocessing and scene timing (Table 3).

2 Problem Formulation

For a fire f , let $\mathbf{x}_f^{\text{pre}}, \mathbf{x}_f^{\text{post}} \in \mathbb{R}^{6 \times H \times W}$ be Landsat surface reflectance in bands 2–7 (blue, green, red, near infrared, shortwave infrared 1 and 2) before and after the fire, and let Ω_f be the set of pixels inside the MTBS fire perimeter that are not masked. The label $y_f(i) \in \{0, 1, 2, 3\}$ for $i \in \Omega_f$ is the MTBS severity class (unburned, low, moderate, high). The Normalized Burn Ratio of an image is

$$\text{NBR} = \frac{\rho_{\text{NIR}} - \rho_{\text{SWIR2}}}{\rho_{\text{NIR}} + \rho_{\text{SWIR2}}}, \quad (1)$$

where ρ_{NIR} and ρ_{SWIR2} are surface reflectance in the near-infrared and second shortwave-infrared bands (OLI bands 5 and 7). The differenced index is $\text{dNBR}_f = 1000(\text{NBR}_f^{\text{pre}} - \text{NBR}_f^{\text{post}})$, scaled by 1000 as in MTBS and Key and Benson [5]. MTBS produces labels approximately as

$$y_f(i) \approx q(\text{dNBR}_f(i); \boldsymbol{\tau}_f), \quad q(v; \boldsymbol{\tau}) = \sum_{k=1}^3 \mathbb{I}[v \geq \tau_k], \quad (2)$$

where $\boldsymbol{\tau}_f = (\tau_{f,1}, \tau_{f,2}, \tau_{f,3})$ are thresholds chosen by the analyst for fire f . The approximation reflects manual edits and differences between the reflectance processing used here and that of MTBS.

Given a set of training fires $\mathcal{F}_{\text{train}}$ with labels, the task is to predict y_{f^*} for a fire $f^* \notin \mathcal{F}_{\text{train}}$ from $\mathbf{x}_{f^*}^{\text{pre}}$ and $\mathbf{x}_{f^*}^{\text{post}}$ alone. Equation (2) suggests two strategies. One estimates thresholds that transfer across fires. The other learns a mapping from reflectance to severity that does not depend on $\boldsymbol{\tau}_{f^*}$. A predictor that knew $\boldsymbol{\tau}_{f^*}$ would reproduce the analyst’s rule up to manual edits, and it serves as an oracle.

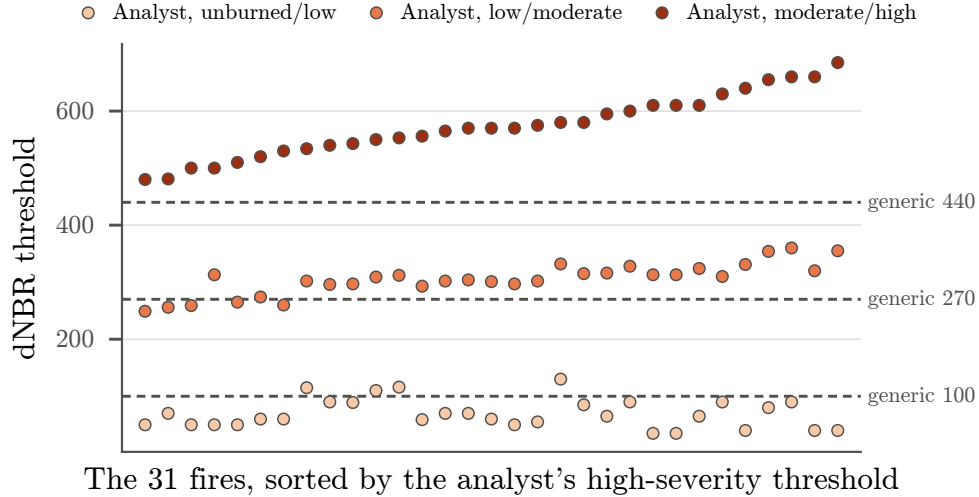


Figure 2: MTBS analyst dNBR thresholds for the 31 fires. Each column is one fire, sorted by its moderate/high threshold. Dashed lines mark the generic thresholds (100, 270, 440).

3 Data

3.1 Fire Selection

The dataset contains every wildfire in the MTBS perimeter database that (i) started in California or Oregon in 2018, 2020, or 2021, (ii) burned at least 60,000 acres, (iii) was mapped with an extended assessment, in which the post-fire scene is acquired in the growing season after the fire, and (iv) used Landsat 8 or 9 for both scenes. These criteria yield 31 fires (Table 4 in Appendix A) covering 7.1 million acres, from Sierra Nevada mixed conifer to the Klamath Mountains, the western Oregon Cascades, the Coast Ranges, and southern California chaparral. The number of labeled pixels per fire ranges from 0.28 to 4.80 million.

3.2 Imagery

The pre- and post-fire scenes identified in each MTBS record are read from the Landsat Collection 2 Level-2 archive [16] hosted on Microsoft Planetary Computer [7]. For 2 fires the archive does not contain the exact post-fire scene, and the closest Landsat 9 scene of the same path and row, acquired 8 days later, replaces it. Digital numbers are converted to surface reflectance with the Collection 2 scale factor and offset ($\rho = 2.75 \times 10^{-5} \text{DN} - 0.2$). Each band is warped with bilinear resampling onto the 30 m Albers grid of the MTBS severity mosaic, which leaves the labels on their native grid. Pixels that the Level-2 quality band flags as fill, cloud, dilated cloud, cirrus, cloud shadow, or snow in either scene are masked. To check alignment, each fire's analyst thresholds are applied to the dNBR computed here and compared with the MTBS map under one-pixel shifts. For 5 of 31 fires (3 of the 7 fires from 2018 and 0 from 2021), agreement is highest at a one-pixel shift, by 0.1 to 1.9 percentage points. The offset is not corrected, and it affects every method equally.

3.3 Labels

Labels come from the MTBS thematic burn severity mosaics for 2018, 2020, and 2021 [9], restricted to each fire's perimeter so that neighboring fires in the mosaic are excluded. The six MTBS classes are unburned to low, low, moderate, high, increased greenness, and a non-processing mask. Unburned to low and increased greenness map to *unburned*, low, moderate, and high are kept, and the non-processing mask is ignored. The 31.4 million labeled pixels are 15.2% unburned, 32.3% low, 28.4% moderate, and 24.1% high. The share of high severity varies from 2% to 47% across fires.

3.4 Analyst Thresholds

Each MTBS record stores the analyst’s three dNBR thresholds. Across the 31 fires the unburned/low threshold ranges from 35 to 130, the low/moderate threshold from 249 to 360, and the moderate/high threshold from 480 to 685 (Figure 2). The generic values of 100, 270, and 440 follow the class edges of Key and Benson [5]. In their scheme, 440–660 is a moderate-high class and values above 660 are high.

3.5 Cross-Validation by Fire

Three pairs of fires overlap. The 2020 North Complex burned into the 2018 Camp Fire scar, the 2021 McFarland Fire into the 2020 August Complex, and the 2021 Bootleg Fire into the 2018 Watson Creek Fire. Fires that share more than 1% of the smaller fire’s area are placed in the same group, and a greedy procedure assigns groups to five folds while balancing burned area. Each fold contains 4 to 8 fires and 1.38 to 1.44 million acres. Every method is trained on four folds and evaluated on the fifth.

4 Methods

Spectral indices. Two relativized indices complement dNBR, the relativized dNBR [8] and the relativized burn ratio [11], both scaled by 1000,

$$\text{RdNBR} = \frac{\text{dNBR}}{\sqrt{\max(|\text{NBR}^{\text{pre}}|, 0.001)}}, \quad \text{RBR} = \frac{\text{dNBR}}{\text{NBR}^{\text{pre}} + 1.001}. \quad (3)$$

Both divide dNBR by a function of pre-fire NBR, which reduces the dependence of dNBR on pre-fire biomass. The floor of 0.001 in the RdNBR denominator and the offset of 1.001 in the RBR denominator prevent division by values near zero.

Index thresholding. Each threshold classifier applies $q(\cdot; \tau)$ from Equation (2) to one index. The generic classifier applies dNBR with $\tau = (100, 270, 440)$. The learned classifier uses thresholds that maximize the mean mIoU over training fires,

$$\hat{\tau} = \arg \max_{\tau} \frac{1}{|\mathcal{F}_{\text{train}}|} \sum_{f \in \mathcal{F}_{\text{train}}} \text{mIoU}_f(q(\text{index}_f; \tau)), \quad (4)$$

found by coordinate search over integer thresholds with step sizes 64, 16, 4, and 1, using per-fire histograms of the index by class. Weighting fires equally prevents the largest fires from determining the thresholds. The oracle applies dNBR with the test fire’s own analyst thresholds τ_{f^*} . It uses information from the test fire and serves as a reference ceiling, not a baseline.

Gaussian smoothing and multi-Otsu. An unsupervised baseline smooths dNBR with a Gaussian filter ($\sigma = 1.5$ pixels) and splits the smoothed values inside the perimeter of each fire into four classes by multi-level Otsu thresholding [10]. It requires no training.

Per-pixel gradient boosting. A histogram-based gradient-boosting classifier [4, 13] uses 300 iterations, 63 leaves per tree, and class weights inversely proportional to class frequency. Each pixel has 17 features, the 12 reflectance values together with NBR^{pre} , NBR^{post} , dNBR, RdNBR, and RBR. Training uses 60,000 labeled pixels sampled uniformly from each training fire. The classifier has no spatial context.

U-Net. The U-Net [14] has four encoder stages of 32, 64, 128, and 256 channels, a 512-channel bottleneck, transposed-convolution upsampling, and skip connections (7.8M parameters). Two variants differ in their inputs. One uses the 12 reflectance bands (*12 bands*). The other adds the five index channels (*12 bands + indices*), with dNBR, RdNBR, and RBR divided by 1000 and RdNBR clipped to $[-5, 5]$. Each channel is standardized with the mean and standard deviation of valid pixels in the training fires. Masked pixels are set to zero after standardization. Training samples 256×256 patches centered on labeled pixels, selects

Table 1: Results on 31 held-out fires (fire-grouped five-fold cross-validation, MTBS labels). mIoU, macro-F1 (F1), and pixel accuracy (Acc.) are averaged over fires ($\pm$ one standard deviation across fires). Pooled is the mIoU of the summed confusion matrices. Per-class IoU is averaged over fires, and U, L, M, and H denote unburned, low, moderate, and high. Bold marks the best non-oracle value. The oracle applies each test fire’s own MTBS analyst thresholds and is a reference ceiling.

Method	Mean over fires / pooled				IoU per class			
	mIoU	F1	Acc.	Pooled	U	L	M	H
Gaussian + multi-Otsu	0.456 $\pm$ 0.102	0.611	0.612	0.464	0.441	0.389	0.402	0.592
dNBR, generic thresholds	0.564 $\pm$ 0.069	0.708	0.718	0.588	0.660	0.609	0.385	0.599
dNBR, learned thresholds	0.684 $\pm$ 0.058	0.808	0.820	0.709	0.662	0.700	0.633	0.743
RdNBR, learned thresholds	0.547 $\pm$ 0.122	0.693	0.700	0.564	0.571	0.552	0.475	0.588
RBR, learned thresholds	0.662 $\pm$ 0.060	0.793	0.802	0.683	0.644	0.675	0.608	0.722
Per-pixel gradient boosting	0.678 $\pm$ 0.060	0.804	0.814	0.705	0.655	0.676	0.639	0.742
U-Net, 12 bands	0.684 $\pm$ 0.058	0.809	0.819	0.708	0.664	0.690	0.636	0.745
U-Net, 12 bands + indices	0.686 $\pm$ 0.057	0.810	0.822	0.711	0.664	0.695	0.642	0.745
Oracle (analyst thresholds)	0.735 $\pm$ 0.043	0.845	0.854	0.752	0.702	0.736	0.693	0.808

a fire with probability proportional to the square root of its number of labeled pixels, and applies random horizontal and vertical flips and 90° rotations. The loss is cross-entropy over labeled pixels, with class weights proportional to the inverse square root of class frequency in the training fires. Training runs for 3,000 steps with batch size 16, using AdamW [6] with weight decay 10^{-4} and a one-cycle learning-rate schedule [15] that peaks at 10^{-3} . The number of steps is fixed in advance, and no checkpoint is selected on held-out data. At test time a 256×256 sliding window at stride 128 covers each fire. A two-dimensional Hann window weights overlapping predictions, and reflection padding ensures that every pixel is predicted. Appendix B lists all hyperparameters.

5 Experimental Setup

Metrics. For a fire f and class c , $\text{IoU}_{f,c} = \text{TP} / (\text{TP} + \text{FP} + \text{FN})$ over the labeled pixels Ω_f . The fire’s mIoU averages $\text{IoU}_{f,c}$ over the classes present in the label or the prediction. Macro-F1 and pixel accuracy are also reported. Two aggregates appear in the results. The mean over fires weights every fire equally, and the pooled mIoU of the summed confusion matrices weights every pixel equally. The two differ because the number of labeled pixels per fire differs by a factor of 17.

Statistical comparison. Two methods A and B are compared by the mean of the per-fire difference $\text{mIoU}_f(A) - \text{mIoU}_f(B)$, with a 95% confidence interval from 10,000 bootstrap resamples of fires [1]. Resampling fires rather than pixels treats the fire as the unit of generalization.

Implementation. All experiments run on one Apple M5 Pro laptop with 64 GB of memory. Index thresholding and gradient boosting run in minutes on the CPU. Each U-Net fold trains in about 22 minutes on the integrated GPU (0.45 seconds per step) with PyTorch [12].

6 Results

Table 1 reports all methods, and Figure 3 shows mean mIoU with confidence intervals.

Threshold calibration. Thresholds learned on the training fires raise mean mIoU from 0.564 to 0.684 (paired difference +0.121, 95% CI +0.094 to +0.146) and improve 29 of 31 fires. The learned thresholds are 77–85, 312–319, and 577–596 across folds (Table 6). They lie close to the analysts’ typical values. The

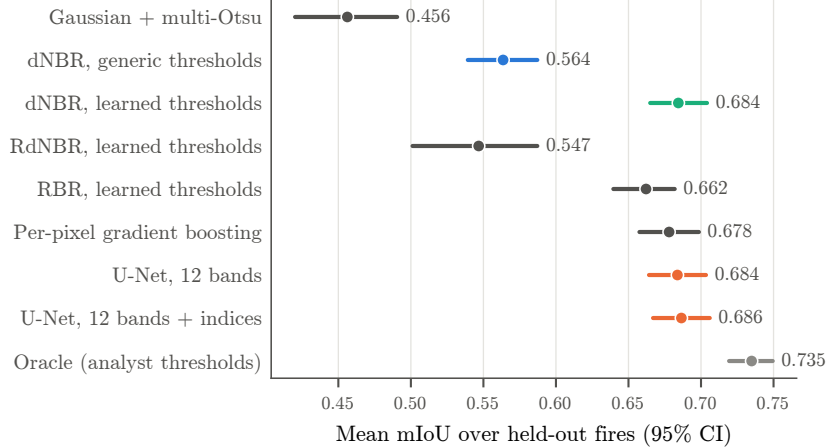


Figure 3: Mean mIoU over the 31 held-out fires with 95% bootstrap confidence intervals from resampling fires.

low/moderate and moderate/high thresholds are well above the generic 270 and 440. The largest per-class change is for moderate severity, whose IoU rises from 0.385 to 0.633.

Deep segmentation. The U-Net on 12 bands reaches 0.684 and the U-Net with index channels reaches 0.686, against 0.684 for dNBR with learned thresholds. The paired differences are -0.001 (95% CI -0.013 to $+0.012$) and $+0.002$ (-0.009 to $+0.014$), and the U-Nets are better on 16 and 15 of 31 fires. Adding index channels to the U-Net changes mIoU by $+0.003$ (-0.004 to $+0.008$). The U-Net with index channels is slightly better than per-pixel gradient boosting ($+0.008$, 95% CI $+0.001$ to $+0.016$), which indicates that spatial context improves on a classifier without it, but not on calibrated dNBR.

Relativized indices. With learned thresholds, RBR reaches 0.662 and RdNBR 0.547. RdNBR divides by the square root of pre-fire NBR, which becomes unstable where pre-fire NBR is near zero. Its standard deviation across fires (0.122) is twice that of dNBR (0.058). Because MTBS labels are defined by dNBR thresholds (Equation (2)), an index other than dNBR is at a disadvantage against these labels regardless of how well it measures field severity.

Unsupervised thresholding. The Gaussian and multi-Otsu pipeline reaches 0.456. Otsu’s method places thresholds that maximize the between-class variance of each fire’s smoothed dNBR histogram, and these need not coincide with severity classes. Its standard deviation across fires (0.102) is the largest of all methods.

Per-fire calibration. The oracle reaches 0.735, $+0.051$ above learned dNBR (95% CI $+0.035$ to $+0.067$), and is better on 25 fires. For each fire, the distance between its analyst thresholds and the learned thresholds of its fold is $\sum_k |\tau_{f,k} - \hat{\tau}_k|$. The oracle’s advantage over learned dNBR increases with this distance (Spearman $\rho = 0.68$, $p < 0.001$, Figure 4). The fires on which learned thresholds fall furthest short are the fires whose analysts departed most from typical thresholds.

Per-fire results. Figure 5 shows mIoU on every fire. Calibrated dNBR and the U-Net track each other closely, and both fall below the oracle on most fires. Figure 6 shows predictions on three held-out fires.

7 Analysis

Bias of generic thresholds. Of the errors made by generic dNBR, 84.4% assign a higher class than MTBS. High severity has 99.3% recall but 63.5% precision, and moderate severity has 48.7% recall, because

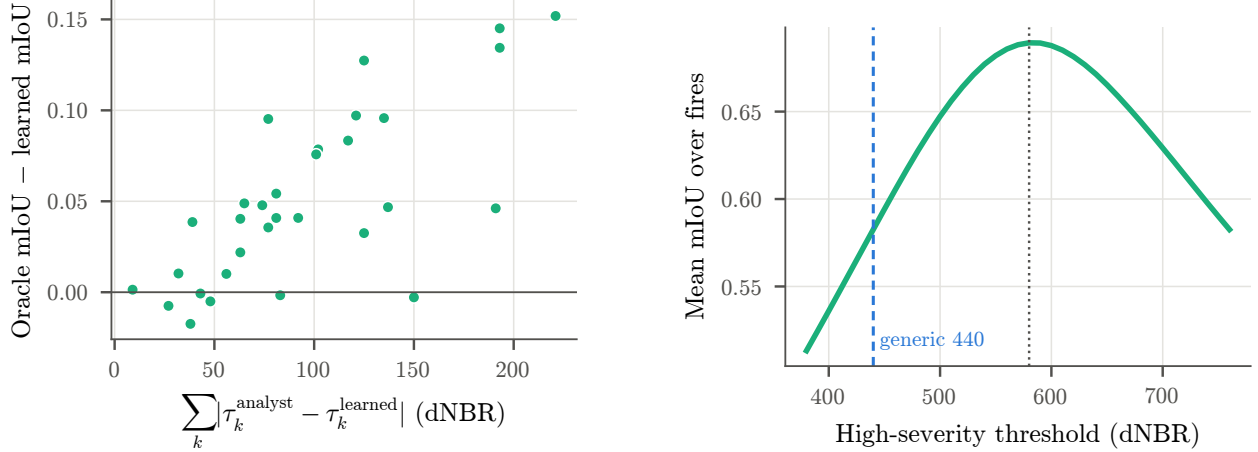


Figure 4: Oracle advantage over learned dNBR on each fire as a function of the distance between its analyst thresholds and the learned thresholds of its fold (left, Spearman $\rho = 0.68$), and mean mIoU over all fires as a function of the moderate/high threshold, with the other two thresholds fixed at their mean learned values of 81 and 314 (right). The right panel is fit on all fires and is descriptive, not a held-out result.

Table 2: Pooled precision (P), recall (R), and F1 (%) per class over all labeled pixels of the 31 held-out fires.

Method	Unburned			Low			Moderate			High		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
dNBR, generic	80.9	86.2	83.5	87.6	67.5	76.2	65.1	48.7	55.7	63.5	99.3	77.5
dNBR, learned	87.3	77.8	82.3	81.0	83.9	82.4	77.4	80.5	78.9	89.1	86.7	87.9
U-Net (17 ch.)	82.0	84.0	83.0	82.5	81.2	81.8	78.7	79.0	78.9	88.3	88.4	88.3
Oracle	91.9	80.2	85.7	84.3	84.6	84.4	80.8	83.1	81.9	89.2	93.0	91.0

the generic moderate/high threshold of 440 is below every analyst’s choice (Table 2). With learned thresholds, moderate recall rises to 80.5% and high precision to 89.1%, and over-predictions fall to 54.3% of errors.

Confusion structure. For each of the four methods in Figure 7, at least 99.1% of errors confuse classes that are adjacent in the severity order. No method confuses unburned with high severity at a measurable rate. The task is therefore one of placing boundaries between neighboring classes, which is the role of thresholds in Equation (2).

Spatial distribution of errors. Each labeled pixel is assigned its distance to the nearest labeled pixel of a different MTBS class. MTBS maps are spatially fragmented, and 57.7% of labeled pixels lie within 30m of another class. The error rate of learned dNBR is 28.0% within 30m of a transition and 0.1% beyond 240m, and 93.2% of its errors lie within 30m of a transition (Figure 8). The U-Net follows the same profile (27.5% and 0.1%). The median distance between a pixel’s dNBR and the nearest analyst threshold is 47 units within 30m of a transition and 199 units beyond 240m, so near transitions a small difference between the learned and the analyst’s thresholds changes the class. The U-Net’s spatial context does not reduce errors at transitions.

Sensitivity to the moderate/high threshold. With the unburned/low and low/moderate thresholds fixed at 81 and 314, mean mIoU over all fires peaks at 0.689 for a moderate/high threshold of 580 and falls to 0.583 at 440 (Figure 4, right). Mean mIoU stays within 0.01 of its maximum for thresholds between 550 and 620, and the generic threshold lies well outside this range.

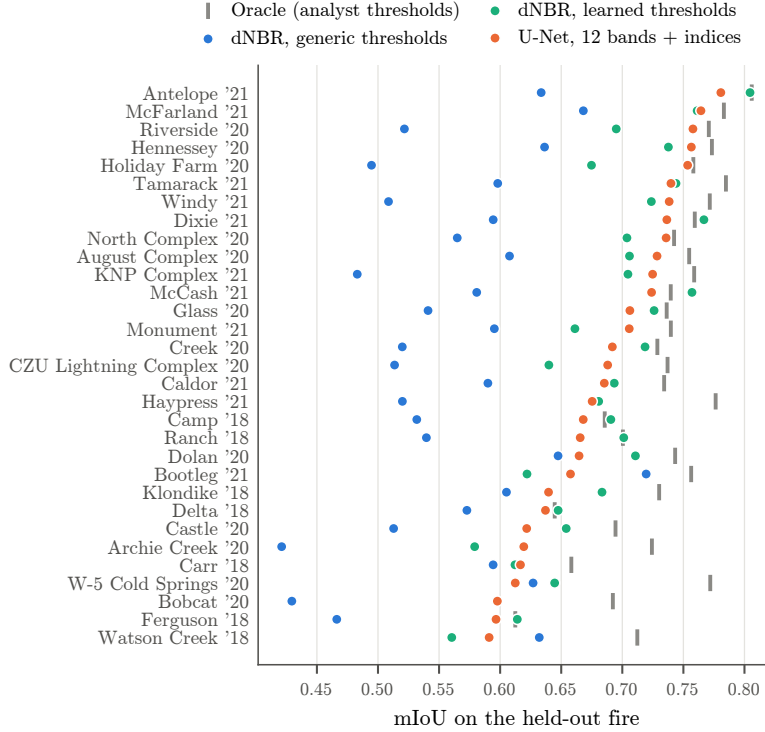


Figure 5: mIoU on each held-out fire, sorted by U-Net score. Gray ticks mark the oracle.

Table 3: Agreement (%) with the MTBS map of the Camp Fire for labels derived by applying the generic thresholds (100, 270, 440) to dNBR, over labeled pixels inside the MTBS perimeter.

Imagery	Pre → post acquisition	Surface reflectance	Percentile stretch
MTBS scenes	Jul 2018 → Jul 2019	70.5	52.1
Landsat 8 median composites	Aug–Nov 2018 → Nov–Dec 2018	43.5	39.9

Label construction. Without analyst maps, burn severity labels can be derived by applying the generic thresholds to dNBR. Such labels encode a fixed rule computed from the model’s own inputs, and they also depend on preprocessing and scene timing (Table 3). For the Camp Fire, generic thresholds applied to surface reflectance from the MTBS scenes agree with the MTBS map on 70.5% of pixels and label 22% of the area as high severity, against 10% in the MTBS map. A per-scene 2nd–98th percentile stretch of each band reduces agreement to 52.1%. Landsat 8 median composites with a post-fire window in November and December 2018, within weeks of the fire rather than in the next growing season, reduce it to 43.5%, and to 39.9% with the stretch.

Training stability. During training of the 12-band U-Net, the loss of one fold became undefined (NaN) at about step 2,600, and that fold’s model predicted every pixel as unburned (mean mIoU 0.034 on its fires). That fold alone was retrained with gradient-norm clipping at 1.0, which gave 0.686. The other nine U-Net folds were trained without clipping and are reported unchanged. The variant with index channels trained stably on every fold.

8 Related Work

Spectral indices for burn severity. The Normalized Burn Ratio contrasts near-infrared reflectance, which healthy vegetation reflects strongly, with shortwave-infrared reflectance, which increases after fire.

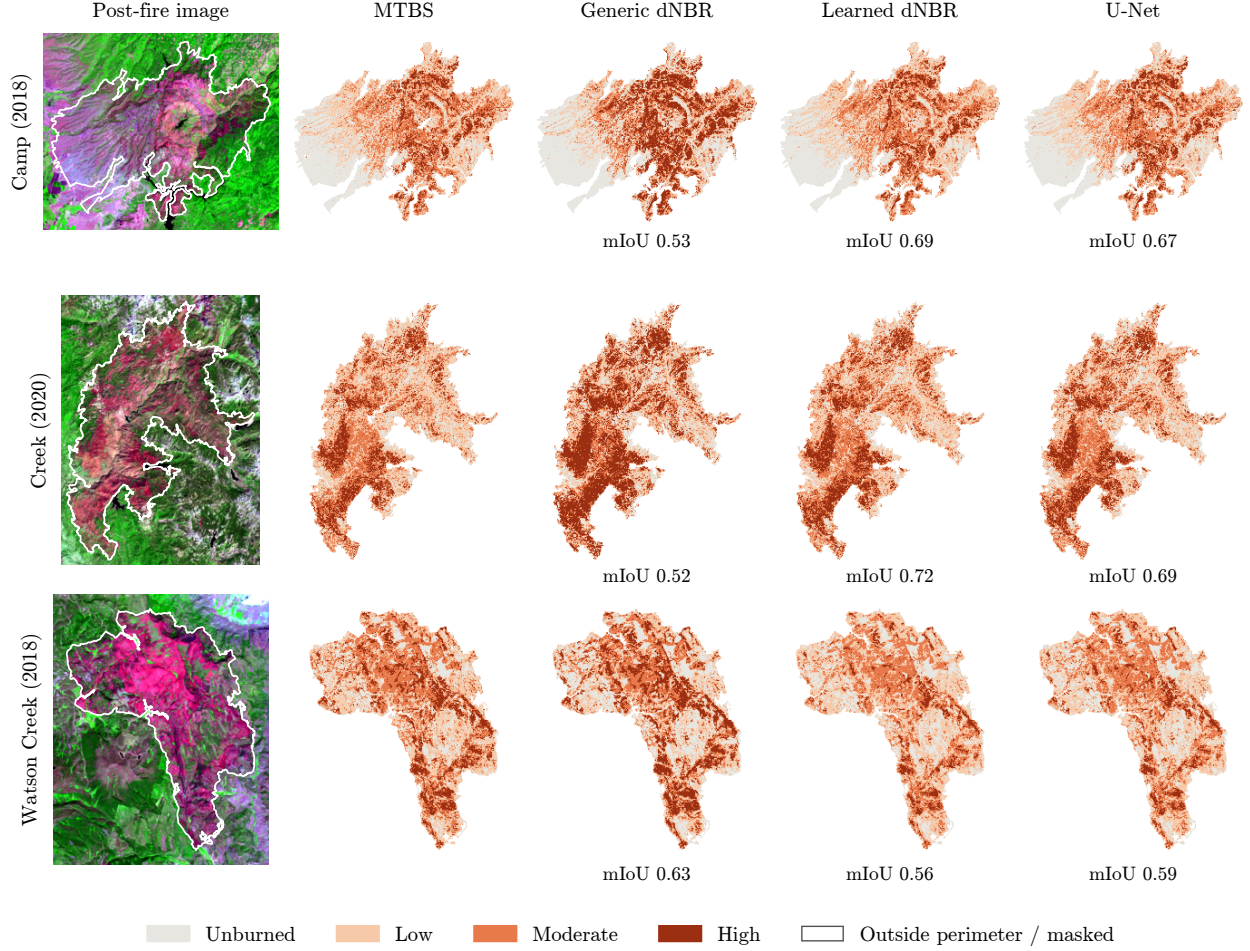


Figure 6: Predictions on three held-out fires, the Camp Fire, the fire with the median U-Net mIoU (Creek), and the fire with the lowest (Watson Creek). Each prediction comes from a model trained on the other folds. The white line is the MTBS perimeter.

Key and Benson [5] defined dNBR and a field protocol, the Composite Burn Index, for calibrating it. Miller and Thode [8] proposed RdNBR to reduce the dependence of dNBR on pre-fire vegetation, and Parks et al. [11] proposed RBR, which corresponded more closely to field measurements across 18 fires than either dNBR or RdNBR. MTBS [2] applies analyst-selected dNBR thresholds fire by fire, which makes its maps both the standard reference and a function of per-fire calibration.

Deep learning for burn severity. U-Net [14] is widely used for dense prediction in remote sensing. Hu et al. [3] trained several U-Net variants on MTBS labels with Landsat imagery across the contiguous United States and found that an attention U-Net performed best among those tested. The comparison here places a U-Net against calibrated index thresholds under an evaluation that holds out whole fires and their spatial neighbors, and it adds an oracle that isolates the effect of per-fire calibration.

Thresholding. Otsu’s method [10] selects thresholds that maximize between-class variance of a histogram and requires no labels. Learned thresholds require labeled fires but target the classes directly.

dNBR, generic					dNBR, learned					U-Net (17 ch.)					Oracle					
MTBS class	U	86	14	0	0	U	78	22	0	0	U	84	16	0	0	U	80	20	0	0
	L	10	67	22	1	L	5	84	11	0	L	9	81	10	0	L	3	85	12	0
	M	0	4	49	48	M	0	11	80	9	M	0	11	79	10	M	0	7	83	10
	H	0	0	1	99	H	0	0	13	87	H	0	0	12	88	H	0	0	7	93
		U	L	M	H		U	L	M	H		U	L	M	H		U	L	M	H
	Predicted					Predicted					Predicted					Predicted				

Figure 7: Pooled confusion matrices over all labeled pixels of the 31 held-out fires, normalized by MTBS class in rows, in percent. U, L, M, and H denote unburned, low, moderate, and high.

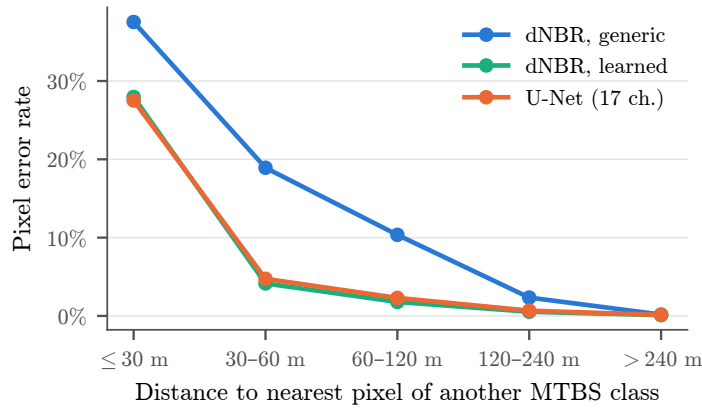


Figure 8: Pixel error rate as a function of distance to the nearest labeled pixel of another class. Learned dNBR and the U-Net overlap. 57.7% of labeled pixels lie in the first bin.

9 Discussion and Limitations

Implications. For reproducing MTBS maps on new fires, a dNBR classifier with thresholds learned from previous fires is as accurate as the U-Net evaluated here, requires no GPU, and has three interpretable parameters. The remaining difference from the analyst is a per-fire calibration problem. A model that predicts a fire’s thresholds from its vegetation, terrain, and imagery targets that difference directly, and the oracle bounds what such a model could gain in this setting.

Reference labels. MTBS maps are derived from dNBR with analyst thresholds and manual edits. They measure agreement with an analyst, not with field severity, and they favor dNBR over other indices by construction. Evaluating against field measurements such as the Composite Burn Index would test burn severity itself.

Scope. The fires come from two states and three years, and fold assignment is fixed. The evaluation covers one U-Net architecture with one training budget, and each fold is trained once. Larger models, longer training, or pretraining on more regions may change the comparison. For two fires the post-fire scene was acquired 8 days after the scene MTBS used, and for 5 fires the imagery and the MTBS labels are offset by up to one pixel (Section 3). The surface-reflectance processing differs from MTBS processing, which contributes to the oracle’s error.

10 Conclusion

Automatic burn severity mapping was evaluated against MTBS analyst maps on 31 wildfires, with whole fires and their spatial neighbors held out. dNBR with thresholds learned on other fires reaches a mean mIoU of 0.684, and two U-Net variants reach the same accuracy. An oracle with each fire’s own analyst thresholds reaches 0.735, and its advantage tracks how far a fire’s thresholds depart from typical values. Remaining errors confuse adjacent classes at class transitions. Under this evaluation, the limiting factor for automatic burn severity mapping is fire-specific threshold calibration.

References

- [1] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993.
- [2] Jeff Eidsenstink, Brian Schwind, Ken Brewer, Zhi-Liang Zhu, Brad Quayle, and Stephen Howard. A project for monitoring trends in burn severity. *Fire Ecology*, 3(1):3–21, 2007.
- [3] Xikun Hu, Puzhao Zhang, and Yifang Ban. Large-scale burn severity mapping in multispectral imagery using deep semantic segmentation models. *ISPRS Journal of Photogrammetry and Remote Sensing*, 196: 228–240, 2023.
- [4] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [5] Carl H. Key and Nathan C. Benson. Landscape assessment (LA): Sampling and analysis methods. In *FIREMON: Fire Effects Monitoring and Inventory System*, General Technical Report RMRS-GTR-164-CD. USDA Forest Service, Rocky Mountain Research Station, Fort Collins, CO, 2006.
- [6] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- [7] Microsoft. Microsoft planetary computer. <https://planetarycomputer.microsoft.com>, 2026.
- [8] Jay D. Miller and Andrea E. Thode. Quantifying burn severity in a heterogeneous landscape with a relative version of the delta normalized burn ratio (dNBR). *Remote Sensing of Environment*, 109(1): 66–80, 2007.
- [9] MTBS Project (USDA Forest Service and U.S. Geological Survey). Monitoring trends in burn severity thematic burn severity mosaics for CONUS, 2018, 2020, and 2021 (ver. 9.0). ScienceBase, <https://www.sciencebase.gov>, 2024.
- [10] Nobuyuki Otsu. A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1):62–66, 1979.
- [11] Sean A. Parks, Gregory K. Dillon, and Carol Miller. A new metric for quantifying burn severity: The relativized burn ratio. *Remote Sensing*, 6(3):1827–1844, 2014.
- [12] Adam Paszke et al. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [13] Fabian Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [14] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241, 2015.

- [15] Leslie N. Smith and Nicholay Topin. Super-convergence: Very fast training of neural networks using large learning rates. In *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications*, volume 11006 of *Proceedings of SPIE*, 2019.
- [16] U.S. Geological Survey. Landsat collection 2 level-2 science products. <https://www.usgs.gov/landsat-missions/landsat-collection-2-level-2-science-products>, 2026.

A Fires

Table 4: The 31 fires in the dataset. Px is the number of labeled pixels inside the MTBS perimeter after masking. U, L, M, and H give the share of labeled pixels in each class. Analyst τ lists the MTBS dNBR thresholds (unburned/low, low/moderate, moderate/high). Scenes are Landsat Collection 2 Level-2 (sensor, acquisition date). [†]The archive does not contain the exact MTBS post-fire scene, and the closest Landsat 9 scene of the same path and row, acquired 8 days later, replaces it.

Fire	Year	Fold	Acres (k)	Px (M)	U / L / M / H (%)	Analyst τ	Pre-fire scene	Post-fire scene
August Complex	2020	0	1068	4.80	17 / 29 / 27 / 27	110 / 309 / 550	LC08 2020-07-15	LC08 2021-07-18
McFarland	2021	0	121	0.55	12 / 40 / 29 / 20	89 / 297 / 543	LC08 2021-07-18	LC08 2022-07-21
Windy	2021	0	98	0.44	6 / 33 / 33 / 28	35 / 313 / 610	LC08 2021-06-11	LC08 2022-06-14
CZU Lightning	2020	0	86	0.38	9 / 31 / 26 / 34	80 / 354 / 655	LC08 2020-07-24	LC08 2021-07-11
Glass	2020	0	68	0.30	15 / 31 / 31 / 23	65 / 316 / 595	LC08 2020-07-15	LC08 2021-07-18
Dixie [†]	2021	1	980	4.37	13 / 32 / 30 / 25	70 / 304 / 570	LC08 2020-07-08	LC09 2022-07-22
Castle	2020	1	174	0.78	15 / 39 / 33 / 13	90 / 310 / 630	LC08 2020-08-04	LC08 2021-08-07
Archie Creek	2020	1	132	0.59	3 / 28 / 29 / 39	40 / 355 / 685	LC08 2020-07-22	LC08 2021-07-25
KNP Complex	2021	1	90	0.40	10 / 43 / 33 / 14	35 / 313 / 610	LC08 2021-06-11	LC08 2022-06-14
Bootleg [†]	2021	2	413	1.86	25 / 23 / 29 / 24	70 / 256 / 481	LC08 2021-06-25	LC09 2022-07-22
Carr	2018	2	234	1.03	4 / 26 / 35 / 35	50 / 259 / 500	LC08 2018-07-10	LC08 2019-07-29
Caldor	2021	2	227	1.02	18 / 27 / 24 / 31	116 / 312 / 553	LC08 2021-08-05	LC08 2022-07-23
Klondike	2018	2	178	0.62	12 / 36 / 22 / 30	50 / 313 / 500	LC08 2018-07-01	LC08 2019-07-20
Riverside	2020	2	138	0.62	13 / 30 / 20 / 37	40 / 331 / 640	LC08 2019-07-20	LC08 2021-07-25
Ferguson	2018	2	97	0.43	12 / 44 / 35 / 9	50 / 297 / 570	LC08 2018-07-05	LC08 2019-07-08
Tamarack	2021	2	70	0.31	17 / 38 / 33 / 11	59 / 293 / 556	LC08 2020-07-01	LC08 2022-07-07
Watson Creek	2018	2	62	0.28	14 / 37 / 34 / 16	50 / 249 / 480	LC08 2017-08-17	LC08 2019-08-07
North Complex	2020	3	317	1.42	5 / 24 / 25 / 47	90 / 328 / 600	LC08 2020-08-09	LC08 2021-07-11
Hennessey	2020	3	314	1.41	34 / 32 / 24 / 11	90 / 296 / 540	LC08 2020-07-08	LC08 2021-07-11
Monument	2021	3	226	1.02	20 / 34 / 25 / 21	115 / 302 / 534	LC08 2021-07-18	LC08 2022-07-21
Holiday Farm	2020	3	174	0.78	10 / 27 / 30 / 33	40 / 320 / 660	LC08 2019-07-20	LC08 2021-07-25
Camp	2018	3	154	0.69	28 / 36 / 25 / 10	60 / 301 / 570	LC08 2018-07-19	LC08 2019-07-22
Bobcat	2020	3	116	0.52	15 / 41 / 33 / 11	130 / 332 / 580	LC08 2020-07-03	LC08 2021-07-06
W-5 Cold Springs	2020	3	84	0.37	19 / 62 / 17 / 2	60 / 274 / 520	LC08 2020-08-02	LC08 2021-08-05
Ranch	2018	4	427	1.92	14 / 36 / 33 / 17	55 / 302 / 575	LC08 2017-07-23	LC08 2019-08-14
Creek	2020	4	382	1.71	10 / 43 / 30 / 16	65 / 324 / 610	LC08 2020-08-27	LC08 2021-08-30
Haypress	2021	4	204	0.86	16 / 38 / 25 / 21	90 / 360 / 660	LC08 2020-07-15	LC08 2022-07-21
Antelope	2021	4	142	0.64	26 / 30 / 22 / 22	85 / 315 / 580	LC08 2021-07-18	LC08 2022-07-21
Dolan	2020	4	124	0.56	10 / 23 / 33 / 34	60 / 260 / 530	LC08 2020-04-28	LC08 2021-05-01
McCash	2021	4	96	0.43	15 / 37 / 31 / 18	70 / 302 / 565	LC08 2021-07-18	LC09 2022-07-20
Delta	2018	4	64	0.29	4 / 25 / 28 / 43	50 / 265 / 510	LC08 2018-07-10	LC08 2019-07-29

B Implementation Details

Table 5: Hyperparameters.

Setting	Value
<i>U-Net</i>	
Encoder channels	32, 64, 128, 256, bottleneck 512
Parameters	7.8M
Input channels	12 (bands) or 17 (bands + indices)
Patch size, batch size	256×256 , 16
Training steps	3,000, fixed, no early stopping
Optimizer	AdamW, weight decay 10^{-4}
Learning rate	one-cycle, peak 10^{-3} , 5% warm-up
Class weights	$\propto (\text{class frequency})^{-1/2}$, training fires
Augmentation	horizontal and vertical flips, 90° rotations
Fire sampling	$\propto (\text{labeled pixels})^{1/2}$
Inference	sliding window, stride 128, Hann weighting, reflection padding
<i>Gradient boosting</i>	
Iterations, leaves	300, 63
Learning rate	0.1
Class weights	balanced
Training pixels	60,000 per training fire
<i>Threshold search</i>	
Objective	mean mIoU over training fires
Search	coordinate search, steps 64, 16, 4, 1
<i>Gaussian + multi-Otsu</i>	
Gaussian σ	1.5 pixels
Otsu classes	4, fit per fire inside the perimeter

Table 6: Thresholds (unburned/low, low/moderate, moderate/high) fit on the four training folds for each held-out fold.

Held-out fold	dNBR	RdNBR	RBR
0	77 / 312 / 579	123 / 471 / 816	58 / 226 / 390
1	82 / 312 / 577	122 / 468 / 809	59 / 224 / 387
2	85 / 319 / 596	126 / 474 / 827	61 / 229 / 394
3	78 / 314 / 587	112 / 474 / 828	55 / 224 / 393
4	81 / 312 / 582	123 / 471 / 820	59 / 227 / 391

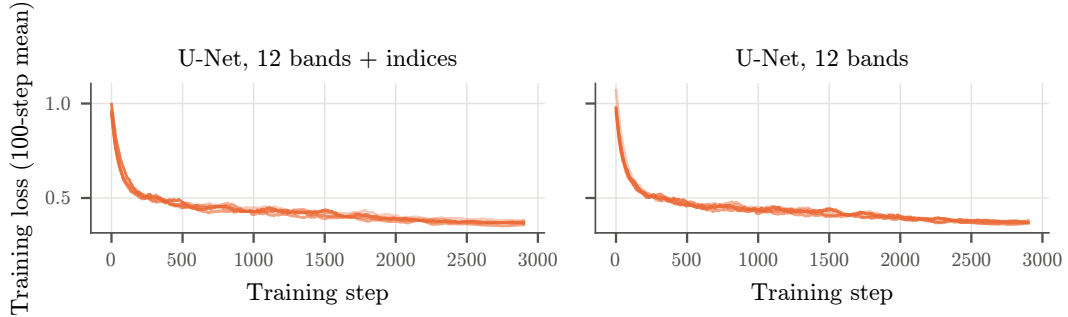


Figure 9: U-Net training loss per fold as a 100-step moving average. Fold 4 of the 12-band variant is the retrained run with gradient clipping.

C Per-Fire Results

Table 7: Per-fire mIoU. Each fire is scored by models trained on the other four folds. Fold is the cross-validation fold in which the fire was tested. Acres in thousands.

Fire	Fold	Acres (k)	Otsu	Generic	Learned	GBM	U-Net	Oracle
August Complex (2020)	0	1068	0.448	0.608	0.706	0.741	0.728	0.755
McFarland (2021)	0	121	0.469	0.668	0.761	0.778	0.764	0.783
Windy (2021)	0	98	0.409	0.509	0.724	0.739	0.738	0.772
CZU Lightning Complex (2020)	0	86	0.442	0.514	0.640	0.657	0.688	0.737
Glass (2020)	0	68	0.522	0.541	0.726	0.693	0.706	0.736
Dixie (2021)	1	980	0.455	0.594	0.767	0.731	0.736	0.759
Castle (2020)	1	174	0.486	0.513	0.654	0.633	0.622	0.694
Archie Creek (2020)	1	132	0.370	0.421	0.579	0.579	0.619	0.724
KNP Complex (2021)	1	90	0.475	0.483	0.705	0.713	0.725	0.759
Bootleg (2021)	2	413	0.602	0.720	0.622	0.635	0.658	0.756
Carr (2018)	2	234	0.263	0.594	0.612	0.620	0.617	0.658
Caldor (2021)	2	227	0.484	0.590	0.693	0.719	0.685	0.734
Klondike (2018)	2	178	0.389	0.605	0.683	0.635	0.640	0.730
Riverside (2020)	2	138	0.388	0.522	0.695	0.735	0.758	0.771
Ferguson (2018)	2	97	0.491	0.466	0.614	0.576	0.597	0.612
Tamarack (2021)	2	70	0.566	0.598	0.744	0.713	0.740	0.785
Watson Creek (2018)	2	62	0.454	0.632	0.560	0.539	0.591	0.712
North Complex (2020)	3	317	0.278	0.565	0.704	0.727	0.736	0.742
Hennessey (2020)	3	314	0.586	0.636	0.738	0.764	0.757	0.773
Monument (2021)	3	226	0.503	0.595	0.661	0.723	0.706	0.740
Holiday Farm (2020)	3	174	0.450	0.495	0.675	0.714	0.753	0.758
Camp (2018)	3	154	0.227	0.532	0.691	0.653	0.668	0.686
Bobcat (2020)	3	116	0.510	0.429	0.597	0.614	0.598	0.692
W-5 Cold Springs (2020)	3	84	0.516	0.627	0.645	0.647	0.612	0.772
Ranch (2018)	4	427	0.575	0.540	0.701	0.660	0.666	0.700
Creek (2020)	4	382	0.518	0.520	0.719	0.666	0.692	0.729
Haypress (2021)	4	204	0.535	0.520	0.681	0.648	0.675	0.776
Antelope (2021)	4	142	0.621	0.634	0.805	0.768	0.781	0.806
Dolan (2020)	4	124	0.338	0.647	0.711	0.688	0.665	0.743
McCash (2021)	4	96	0.515	0.581	0.757	0.692	0.724	0.740
Delta (2018)	4	64	0.260	0.573	0.647	0.620	0.637	0.645

D Reproducibility

All code is in the project repository under `robust/`. The script `fires.py` selects fires and folds from the MTBS perimeter database, and `fetch.py` downloads Landsat scenes and MTBS labels. The scripts `baselines.py` and `unet.py` train and evaluate all methods. The scripts `report.py` and `analysis.py` generate every table, figure, and number in the text.